

Example 12.1

$$X_t = Z_t - \frac{7}{12} Z_{t-1} + \frac{1}{12} Z_{t-2}$$

$$= \sum_{j=0}^{\infty} b_j Z_{t-j}$$

where $b_0 = 1$; $b_1 = -\frac{7}{12}$; $b_2 = \frac{1}{12}$; $b_j = 0$ $j > 2$.

$$B(\xi) = \sum_{j=0}^{\infty} b_j \xi^j$$

$$= 1 - \frac{7}{12} \xi + \frac{1}{12} \xi^2$$

$$= (1 - \delta_1 \xi)(1 - \delta_2 \xi)$$

$$= \left(1 - \frac{1}{3} \xi\right) \left(1 - \frac{1}{4} \xi\right)$$

$$\delta_1 = \frac{1}{3}; \quad \delta_2 = \frac{1}{4}.$$

$$\frac{1}{B(\xi)} = \frac{1}{\left(1 - \frac{1}{3} \xi\right) \left(1 - \frac{1}{4} \xi\right)}$$

$$= \frac{A_1}{1 - \frac{1}{3} \xi} + \frac{A_2}{1 - \frac{1}{4} \xi} \quad \left] \text{Partial fractions} \right.$$

$$(a) \quad k = 1$$

$$D_1(\xi) = \frac{\sum_{j=0}^{\infty} b_{j+1} \xi^j}{B(\xi)}$$

$$= \frac{b_1 + b_2 \xi}{B(\xi)}$$

$$d_j(1) = \begin{cases} \theta_1 & j=0 \\ \frac{1}{\delta_1 - \delta_2} [\theta_1 (\delta_1^{j+1} - \delta_2^{j+1}) + \theta_2 (\delta_1^j - \delta_2^j)] & j \geq 1 \end{cases}$$

$$\hat{X}_{n+1}(1) = \sum_{j=0}^{\infty} d_j(1) X_{n-j}$$

$$= -\frac{7}{12} X_n$$

$$+ \frac{1}{\left(\frac{1}{3} - \frac{1}{4}\right)} \sum_{j=1}^{\infty} \left\{ \left(-\frac{7}{12}\right) \left[\left(\frac{1}{3}\right)^{j+1} - \left(\frac{1}{4}\right)^{j+1} \right] + \left(\frac{1}{12}\right) \left[\left(\frac{1}{3}\right)^j - \left(\frac{1}{4}\right)^j \right] \right\} X_{n-j}$$

(b) $k=2$

$$d_j(2) = \theta_2 \left(\frac{\delta_1^{j+1} - \delta_2^{j+1}}{\delta_1 - \delta_2} \right)$$
$$= \frac{1}{12} \left(\frac{\left(\frac{1}{3}\right)^{j+1} - \left(\frac{1}{4}\right)^{j+1}}{\frac{1}{3} - \frac{1}{4}} \right)$$

$$= \left(\frac{1}{3}\right)^{j+1} - \left(\frac{1}{4}\right)^{j+1}$$

$$\hat{X}_{n+2} = \sum_{j=0}^{\infty} d_j(2)$$

$$= \sum_{j=0}^{\infty} \left[\left(\frac{1}{3}\right)^{j+1} - \left(\frac{1}{4}\right)^{j+1} \right] X_{n-j}$$